

Mapping the Water-Access Evidence Base

A diagnostic review of geographic and methodological gaps in 1,159 AI-extracted studies

Source: Preliminary Results Report | Generated
2026-09-29 | 34,594 initial records

THE CORPUS

1,159

Studies extracted
(85% published
post-2010).

THE GEOGRAPHIC VOID

52%

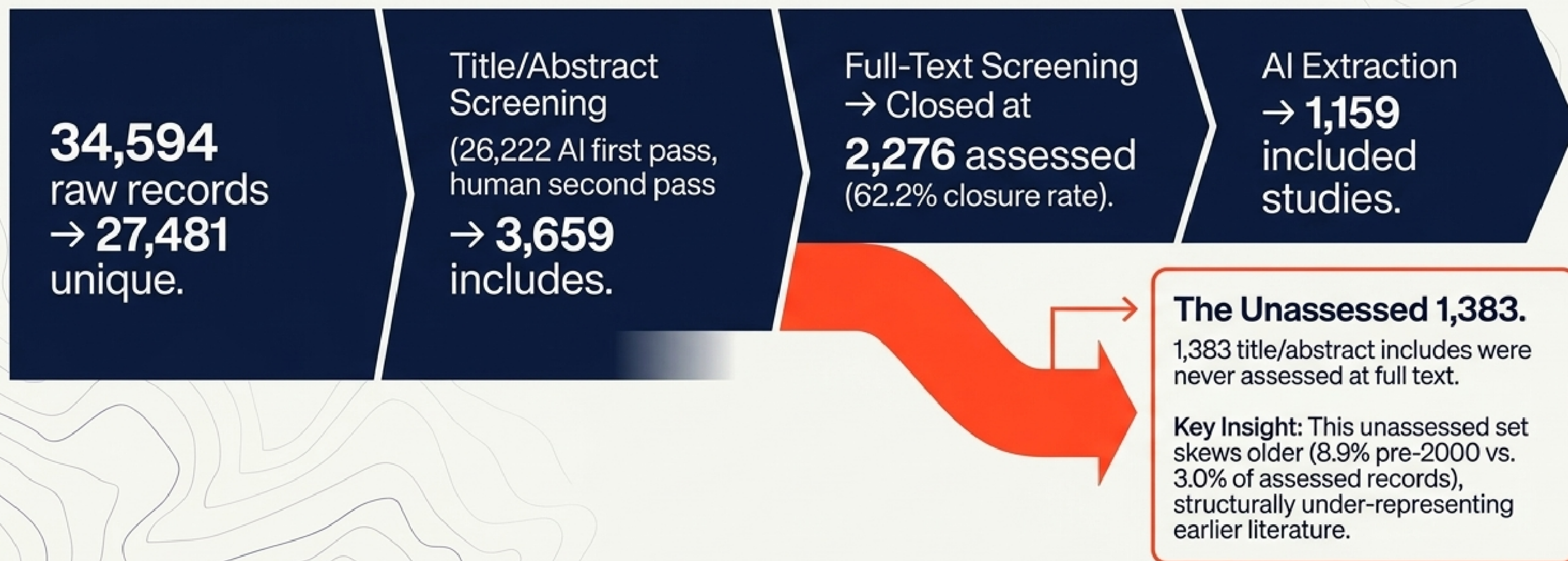
Concentrated in just
10 countries (heavily
Global South). Target
comparison nations
Canada and the
Netherlands are
virtually
unrepresented.

THE METHODOLOGICAL SKEW

6%

Studies capable of
supporting causal
claims. 94% of the
corpus is qualitative,
observational, or
doctrinal.

The search funnel closed early, leaving older literature unassessed



Over half the evidence base sits in ten nations

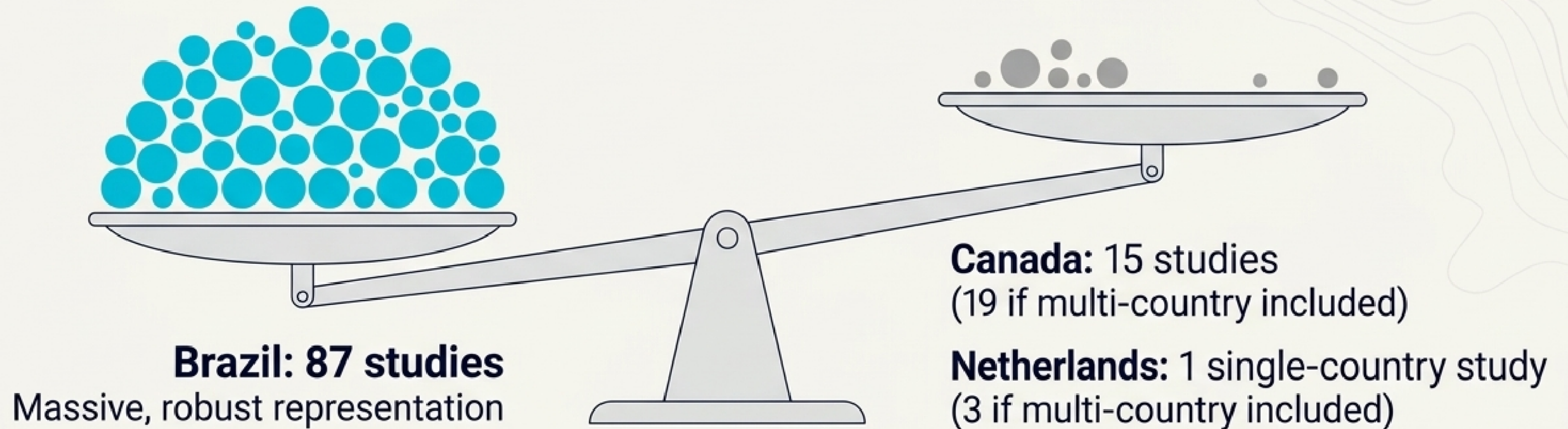


Top Ten Study Locations

India:	133 studies
Brazil:	87 studies
South Africa:	84 studies
USA:	71 studies
Ghana:	53 studies
Kenya:	49 studies
Mexico:	37 studies
Indonesia:	30 studies
Nigeria:	28 studies
Bangladesh:	27 studies

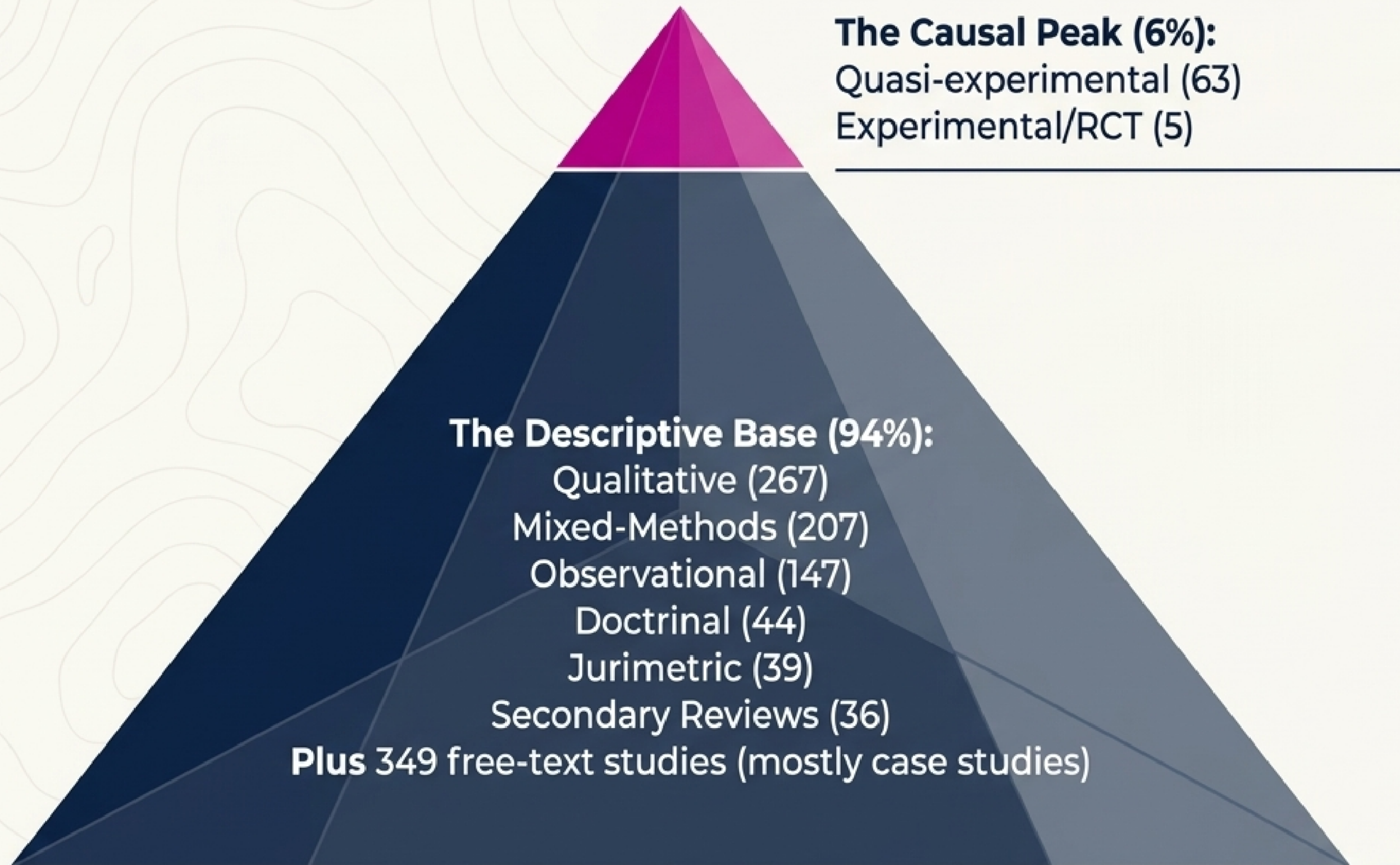
599 studies (52%) are concentrated in these top ten countries. The corpus is overwhelmingly focused on urban and informal settlements (525 studies) and common law systems (546 studies).

The comparative framework lacks foundational evidence in Northern jurisdictions



The dissertation's overarching comparative frame cannot be tested from this base as it currently stands. The evidence base in the Netherlands and Canada is functionally a data desert.

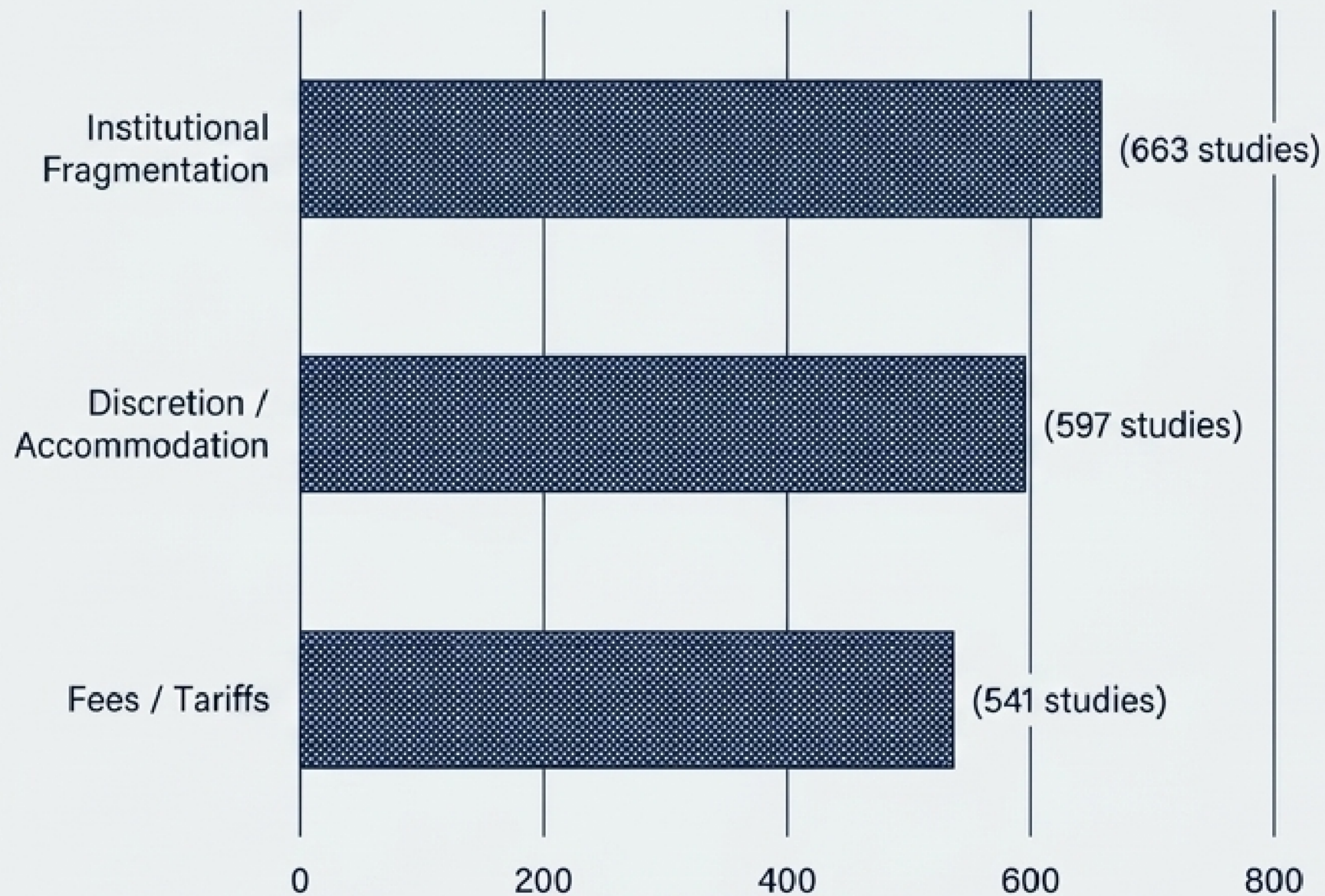
94% of the corpus is qualitative or observational



Appraisal Context:
447 studies (39%) had to be appraised using the project's own non-validated 'Legal Framework' tool because they fit no standard published instrument.

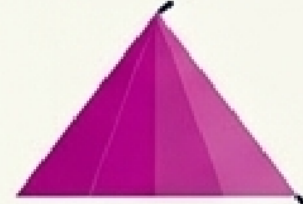
Qualitative research heavily focuses on institutional fragmentation and official discretion

Study Volumes by Mechanism Families

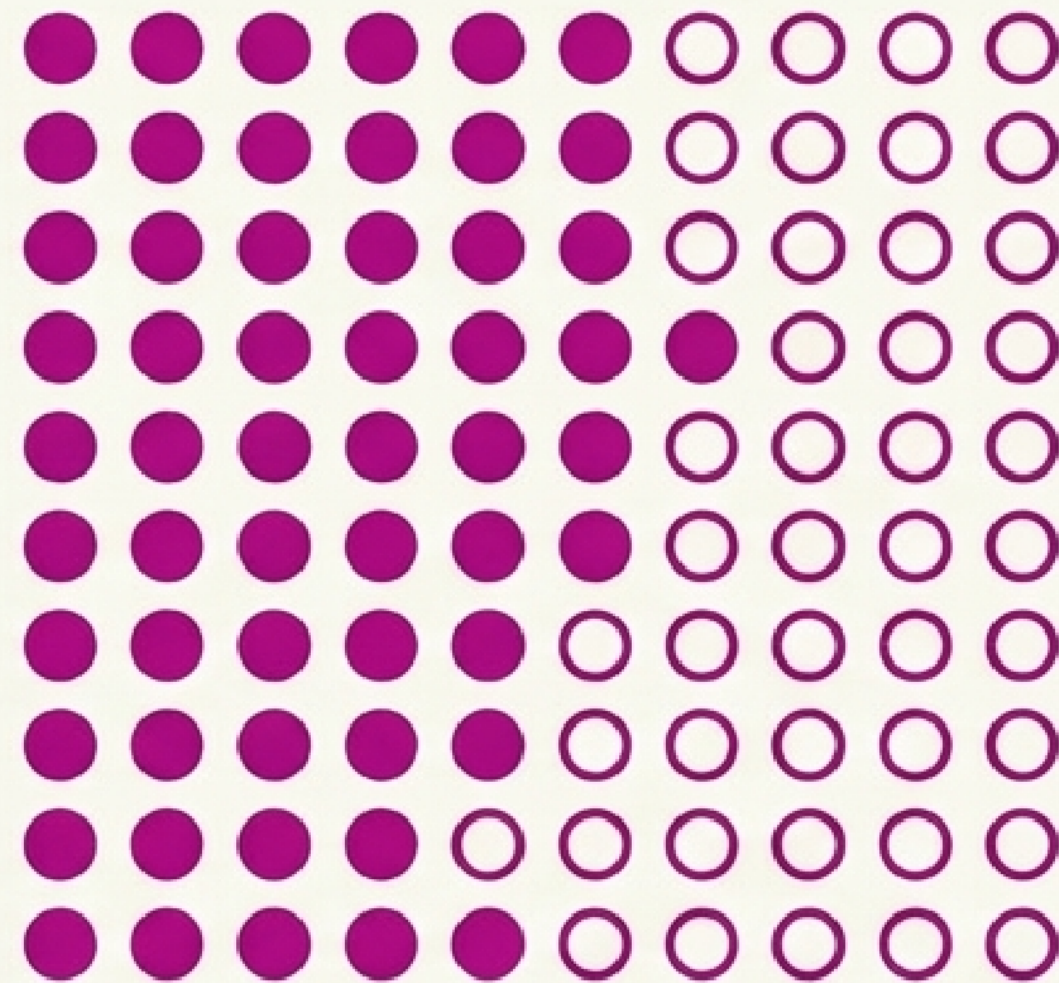


“The literature documents **co-occurrence far more than causation**. The base can say that legal/administrative features **and access outcomes** are associated across many settings, **but rarely** that one causes the other.”

The 68 causal-capable designs offer limited high-certainty data



Causal
Peak



Total Causal Studies: 68
(63 quasi-experimental, 5 RCTs).

Nuance 1: The 5 RCTs actually only represent 4 distinct trials (two papers report on the same DRC cluster trial).

Nuance 2: Certainty Downgrades.
Only 35 of these studies carry a numeric mechanism-certainty of 3 or 4.

High-certainty causal identification regarding legal/administrative mechanisms in the water sector is exceedingly rare.

Evidence clusters around formal connections while ignoring redress and review

The Hot Spots: Water Access (1,013) heavily intersects with Fragmentation (663) and Discretion (597). Formal connection (577) and Affordability (507) are highly researched.

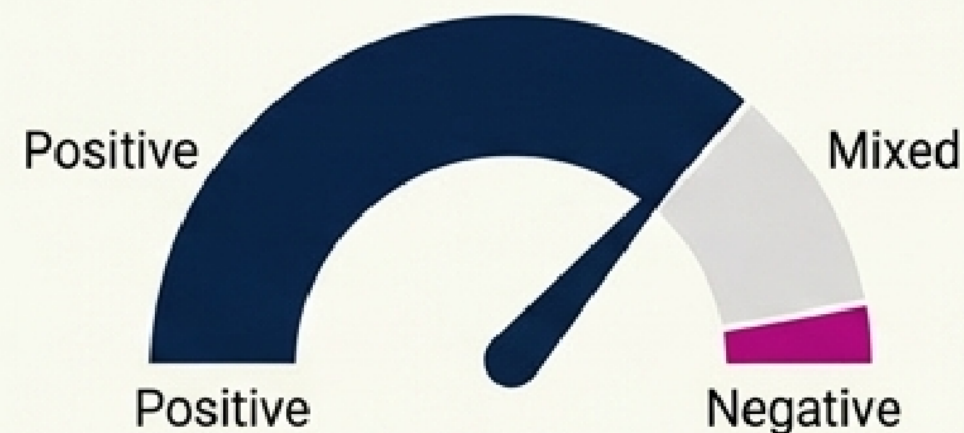
	Water Access	Sanitation Access	Affordability	Formal Connection	Reconnection
Fragmentation	663	457		663	
Discretion	597	457	351	579	
Administrative Review	507		507	577	
Judicial Review	69				
Fees/Tariffs	65	65	23		

The Voids: Sanitation access (457) is studied less than half as often as water. Administrative review (69), Judicial review (65), and Reconnection (23) are almost completely ignored.

Structured syntheses show fragmented agreement on outcomes

A Recognition & Eligibility (20 studies)

15/20 Concordant (75%)



Formal status generally associates with better access, though counter-patterns exist (e.g., zoning status).

B Assistance & Trust (6 studies)

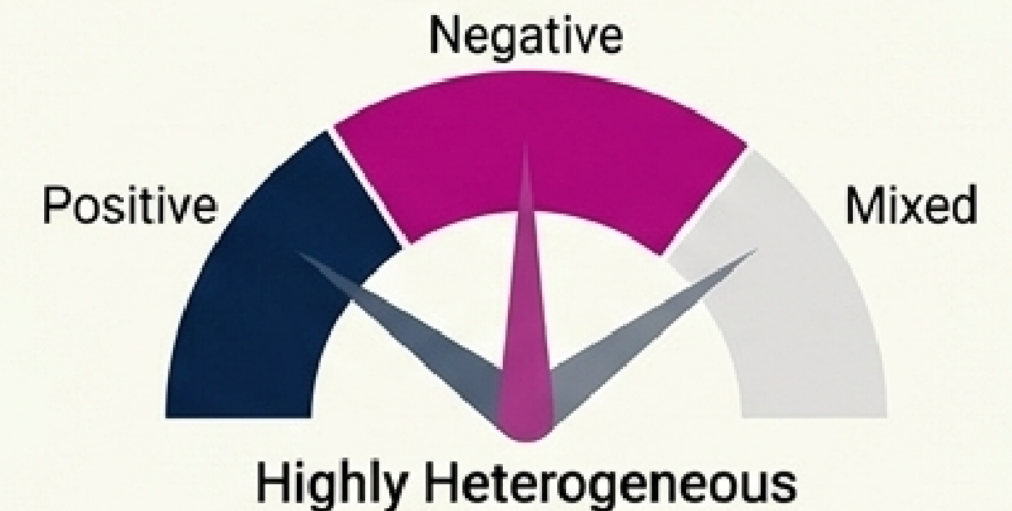
Uniformly Positive/Mixed (0 Negative)



Favorable associations, but drawn from a tiny sample of six different mechanisms.

C Ownership & Price (20 studies)

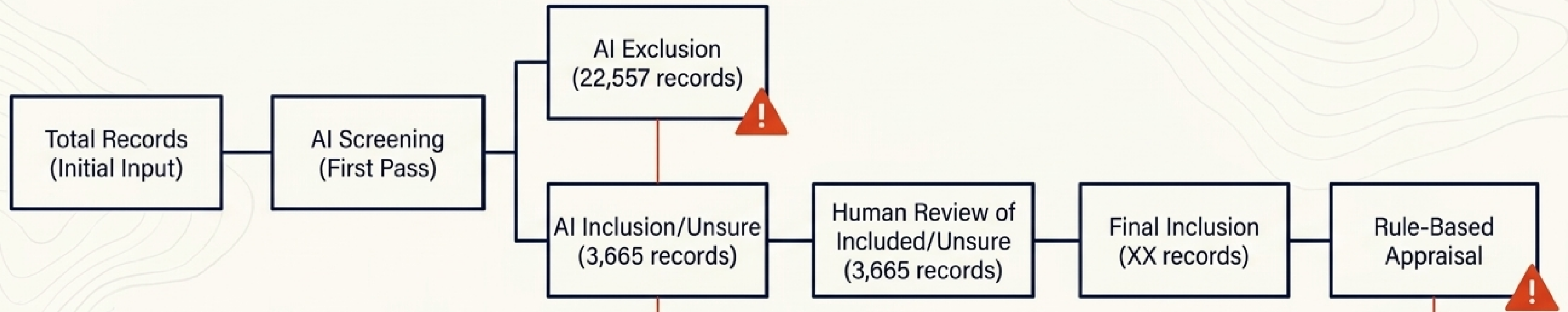
7 Positive / 9 Negative / 4 Mixed



No dominant direction. However, private ownership consistently associates with worse affordability in US/Brazil studies.

Critical blind spots in the AI-assisted screening process

Detailed flowchart the AI vs. Human screening and appraisal process



Warning 1: Unverified Exclusions.

Data: Human review achieved 99.8% agreement on the 3,665 included/unsure records.

The Catch: The 22,557 records excluded by the AI in the first pass received zero human checks. **Any false negatives are permanently invisible.**

Warning 2: Rule-Based Appraisal.

Data: No human review of AI appraisal ratings.

The Catch: 59 of 140 JBI ratings are “High concern”—but this is largely due to sparse AI extraction, not necessarily poor study design. Several items default to “Can’t tell”.

Unresolved tool classifications require human intervention

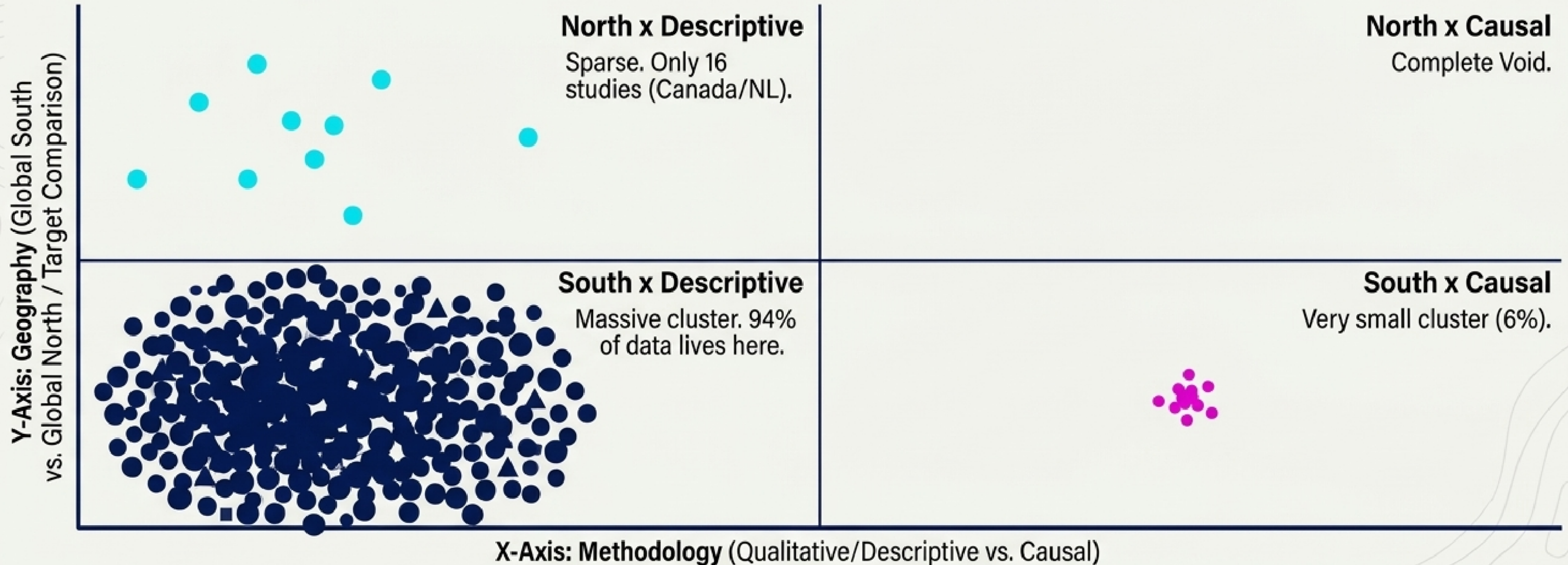
AMSTAR 2 Misalignments (24 Studies)

- └ Only **12** carry a confirmed systematic method label; **12** rely on bare AI classification.
- └ Many are scoping/realist/policy reviews, meaning **AMSTAR 2 (designed for healthcare interventions)** is an **out-of-design** tool.
- └ Result: **22 of 24** are “**Not ratable**” due to missing extraction data.

RoB 2 Cluster RCTs (5 Studies)

- └ **All 5** are confirmed as cluster-randomized (not individually randomized) and were appraised with the cluster variant.
- └ All rate “**Some concerns**” heavily driven by “**No information**” answers from sparse AI extraction.

The Comparative Blind Spot limits structural testing



The current evidence base provides a rich, descriptive map of Global South challenges, but utterly fails to provide the causal, Northern-jurisdiction data required by the dissertation's framing. Primary data collection or highly targeted supplementary searching in Canada and the Netherlands is mandatory.